

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1850

Cajalco Tap 100 kV
July 24, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1850. The Interconnection Customer proposes a 88.9 MW solar photovoltaic facility interconnecting at the Cajalco Tap 100 kV substation in the Illinois area. The request was received on August 06, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 455 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,631,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Guardian - Anza 100 KV	Anza Substation - Sky Valley Substation 100 KV	133.2	123	20.3
Johnson 1 Substation - Shepherds North 100 KV	Cajalco Tap - Rio Linda 100 KV	125.7	224	100.0
Flushing - Rio Linda 100 KV	Shepherds North - Cajalco Tap 100 KV	125.1	228	100.0
Cajalco Tap - Rio Linda 100 KV	Shepherds N - Cajalco Tap 100 KV	124.2	235	100.0
Cajalco Tap - Shepherds North 100 KV Line	Rio Linda - Cajalco Tap 100 KV	124.2	235	100.0
Sky Valley - Johnson 1 100 KV L	Cajalco Tap - Rio Linda 100 KV	120.7	131	66.9
Talentino Sw Sta - Lynn Vly 100 KV	Camel Tracks - Flushing 100 KV	110.0	170	47.0
Camel Tracks Substation - Flushing Substation 100 KV	Talentino Sw Sta - Lynn Valley 100 KV Line	108.2	160	48.2
Boca Raton - Carlyss 230 KV L	Top Crop - Carlyss 500 KV	106.0	724	33.5
Johnson 1 - Sewall 100 KV	Cajalco Tap - Rio Linda 100 KV	103.4	118	33.1
Anza - Sky Vly 100 KV line	Front Range - Shoreline 100 KV Line	100.7	263	28.3
Amrad Substation - Moore Rd 100 KV	Front Range - Shoreline 100 KV line	100.5	231	22.6
Amrad - Talentino Sw Sta 100 KV	Shoreline Substation - FrontRange 100 KV Line	100.3	236	22.6
Anza 230 kV/100 kV XFMR	Shoreline - Front Rnge 100 KV	100.1	420	35.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Anza - Guardian 100 KV	Rebuild line with higher-capacity conductor	\$2,319,000	17% (\$393,000)
Johnson 1 Substation - Shepherds North 100 KV	Rebuild line with higher-capacity conductor	\$25,043,000	28% (\$7,106,000)
Rio Linda - Flushing 100 KV L	Rebuild line with higher-capacity conductor	\$3,262,000	28% (\$926,000)
Cajalco Tap - Rio Linda 100 KV	Rebuild line with higher-capacity conductor	\$5,664,000	28% (\$1,607,000)
Cajalco Tap - Shepherd N 100 KV L	Rebuild line with higher-capacity conductor	\$2,363,000	28% (\$671,000)
Sky Valley - Johnson 1 100 KV L	Rebuild line with higher-capacity conductor	\$7,282,000	27% (\$1,972,000)
Talentino Sw Sta - Lynn Vly 100 KV	Reconductor line	\$2,006,000	24% (\$475,000)
Flushing - Camel Tracks 100 KV	Reconductor line	\$3,109,000	23% (\$704,000)
Boca Raton - Carlyss 230 KV L	Reconductor line	\$19,907,000	18% (\$3,651,000)
Johnson 1 - Sewall 100 KV	Reconductor line	\$4,723,000	26% (\$1,222,000)
Anza - Sky Vly 100 KV	Reconductor line	\$3,938,000	8% (\$320,000)
Moore Rd - Amrad Substation 100 KV	Reconductor line	\$2,106,000	10% (\$218,000)
Amrad - Talentino Sw Sta 100 KV	Reconductor line	\$4,849,000	10% (\$501,000)
Anza Substation 230/100kV Autotransformer	Replace transformer with higher-rated unit	\$12,014,000	7% (\$865,000)
	Total allocated to project		\$20,631,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.